PHYS 210 Electricity and Magnetism

Celebration of Knowledge!

Supplementary Practice 11 - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

Attempt all questions. Supporting practice; this topic mix is not the announced main-test format.

Write all rough calculations and final answers on separate paper. Show your working.

Original source numbering is retained inside each question. Answer the entire cropped question.

---

1. [____] Tutorial 7 - Question 4

4. A 25-μF capacitor charged to 50 V and a capacitor C charged to 20 V are connected to each other, with the two positive plates connected and the two negative plates connected. The final potential difference across the 25-μF capacitor is 36 V. What is the value of the capacitance of C?
   a. 43 μF
   b. 29 μF
   c. 22 μF
   d. 58 μF
   e. 63 μF

2. [____] Tutorial 7 - Question 5

5. A 30-μF capacitor is charged to 80 V and then connected across an initially uncharged capacitor of unknown capacitance C. If the final potential difference across the 30-μF capacitor is 20 V, determine C.
   a. 60 μF
   b. 75 μF
   c. 45 μF
   d. 90 μF
   e. 24 μF

3. [____] Textbook 15e - Question 24.49

24.49 •• A 20.0 μF capacitor is charged to a potential difference of 800 V. The terminals of the charged capacitor are then connected to those of an uncharged 10.0 μF capacitor. Compute (a) the original charge of the system, (b) the final potential difference across each capacitor, (c) the final energy of the system, and (d) the decrease in energy when the capacitors are connected.

4. [____] Textbook 15e - Question 24.57

24.57 •• Three capacitors having capacitances of 8.4, 8.4, and 4.2 μF are connected in series across a 36 V potential difference. (a) What is the charge on the 4.2 μF capacitor? (b) What is the total energy stored in all three capacitors? (c) The capacitors are disconnected from the potential difference without allowing them to discharge. They are then reconnected in parallel with each other, with the positively charged plates connected together. What is the voltage across each capacitor in the parallel combination? (d) What is the total energy now stored in the capacitors?